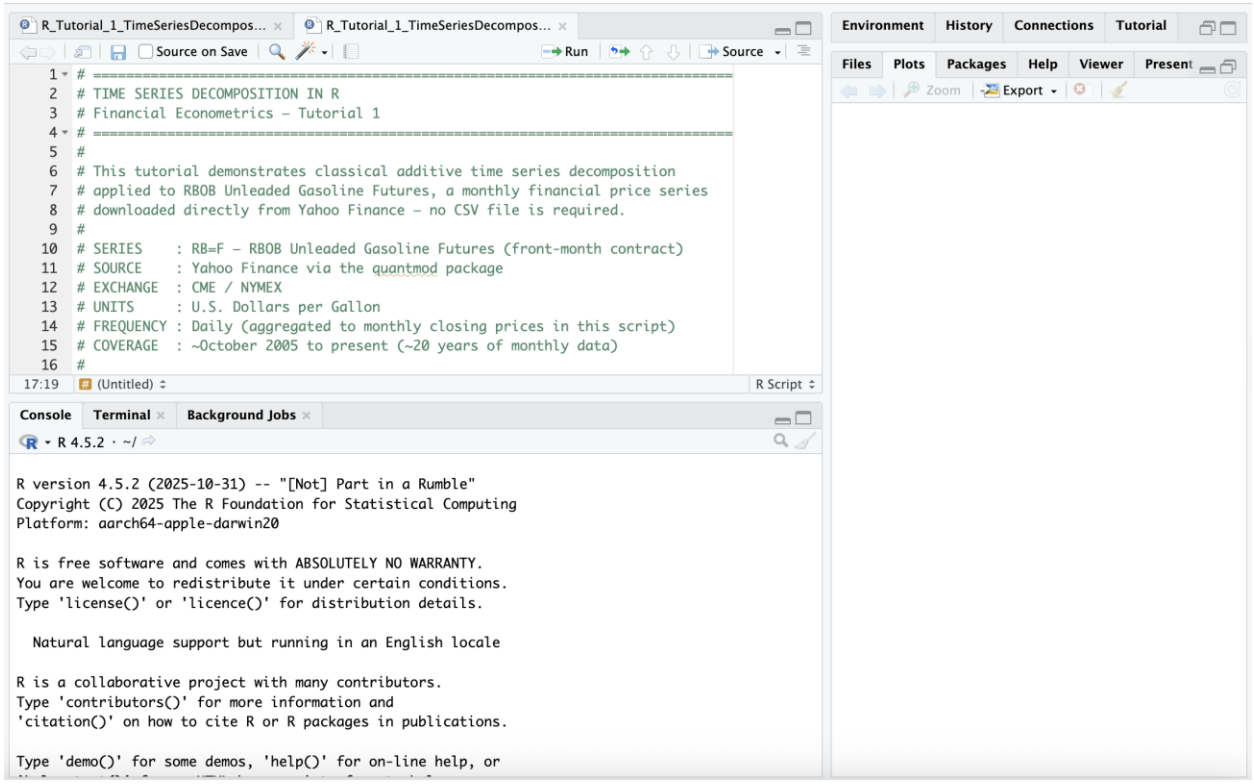
Financial Econometrics, R Tutorial 1

Time Series Decomposition in R

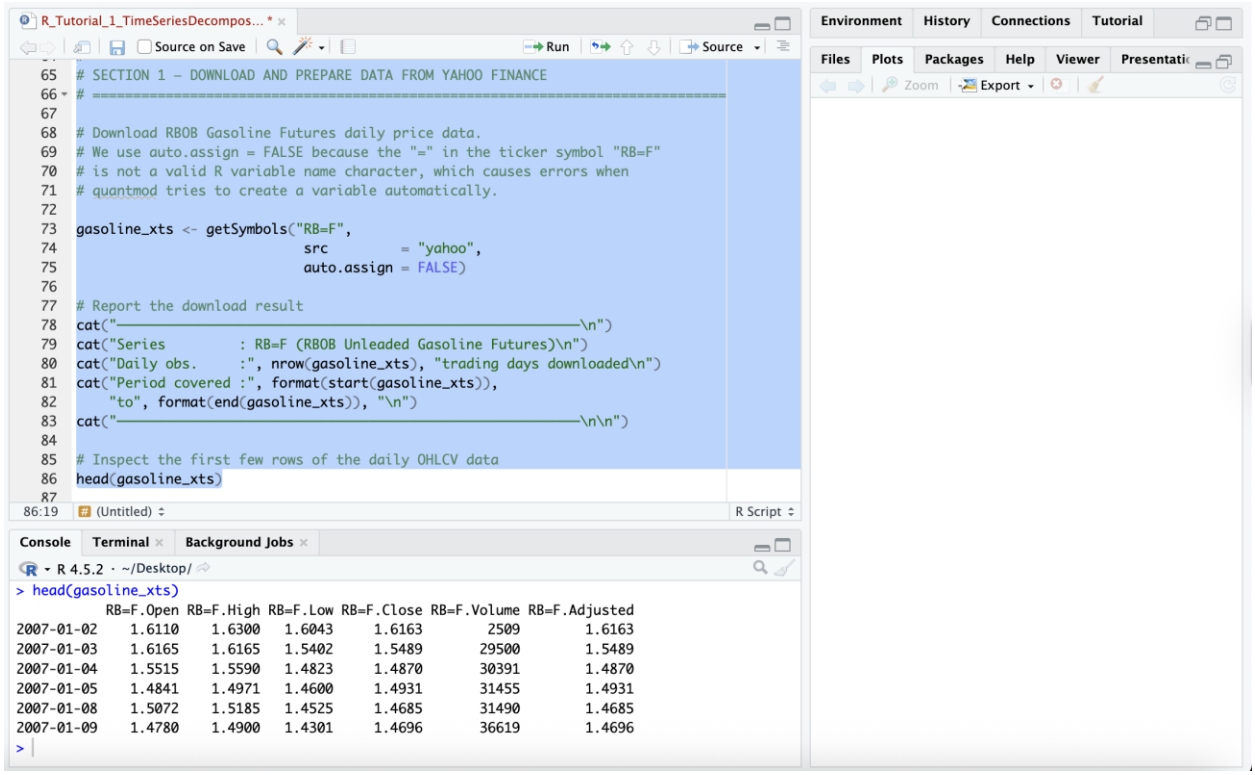
Welcome to this training session. This tutorial focuses on the time series decomposition exercise completed in class. In completing this training, you will use the RBOB Unleaded Gasoline Futures data downloaded directly from Yahoo Finance.

The following is a step-by-step description of the steps required to complete the decomposition.

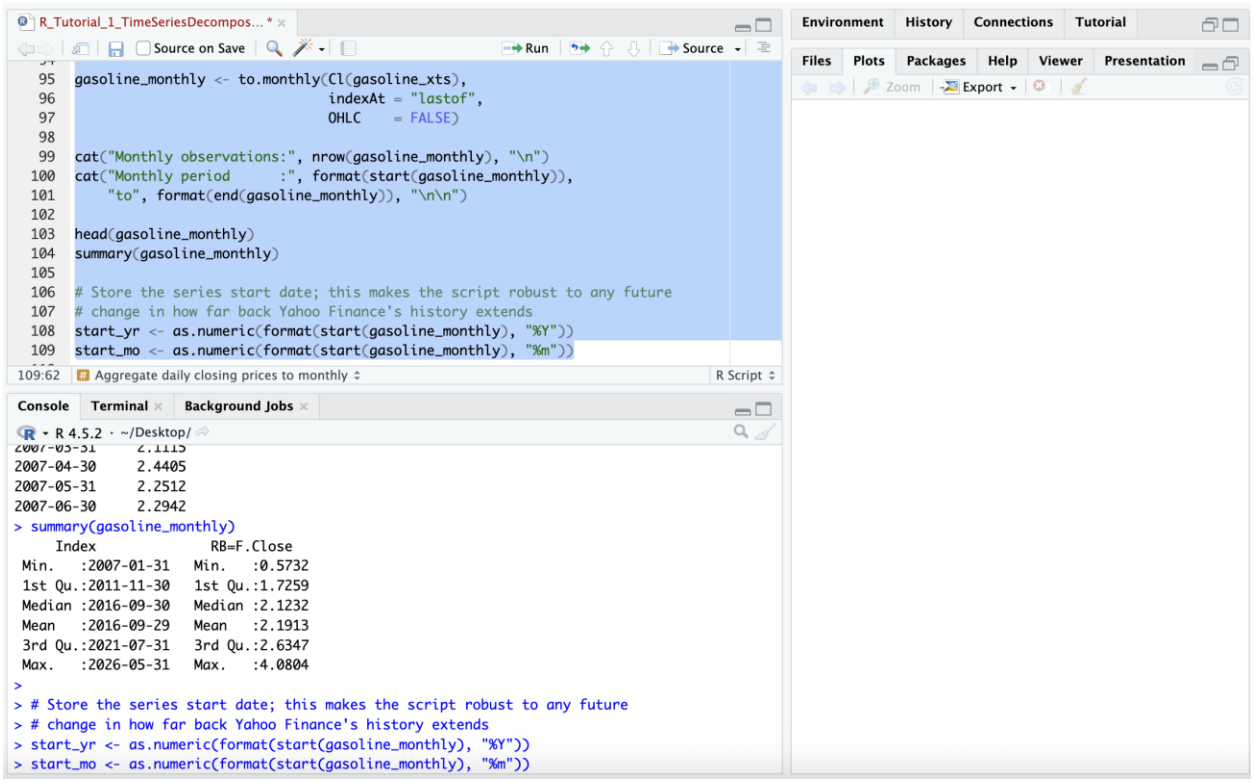
1. Open the R code file called 'R_Tutorial_1_TimeSeriesDecomposition_GasolineFutures' in RStudio. You should see this:



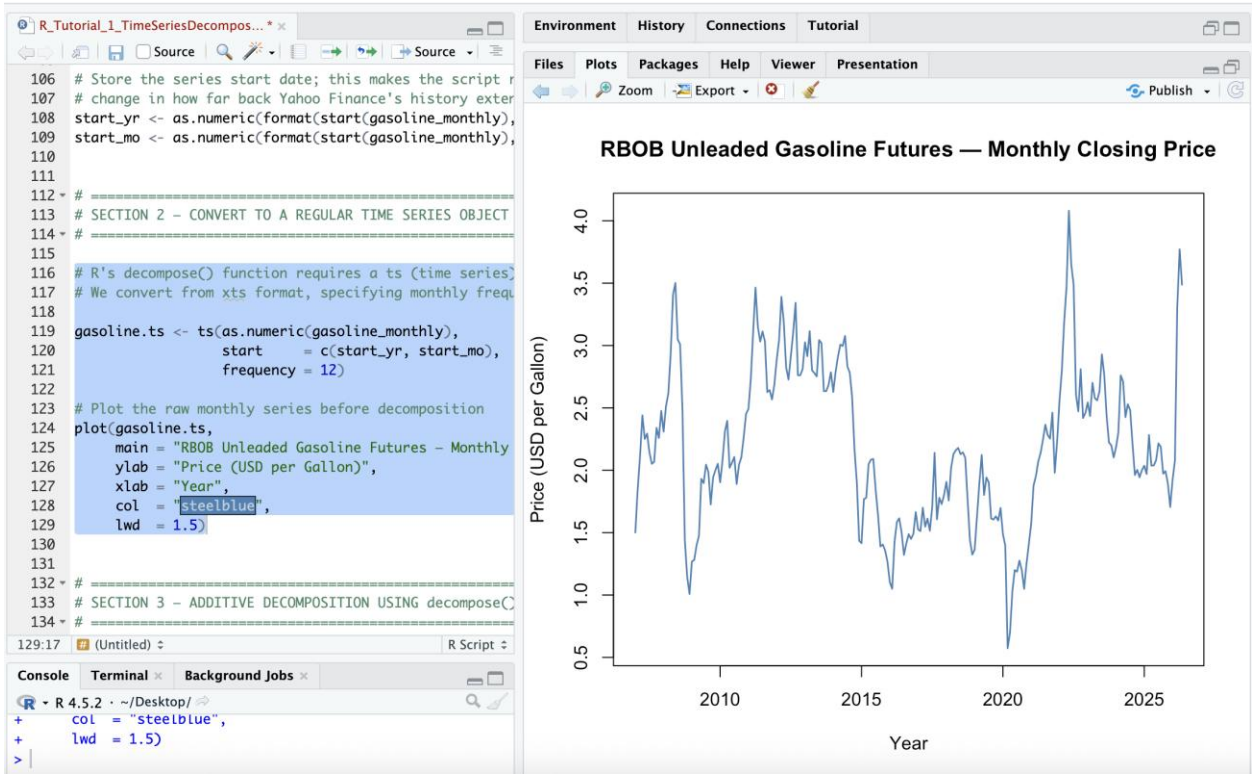
2. Before starting the decomposition, we first need to download financial data directly from Yahoo Finance. Then we need to check that the data was downloaded correctly. For this, run the head function, and you will get the following result:



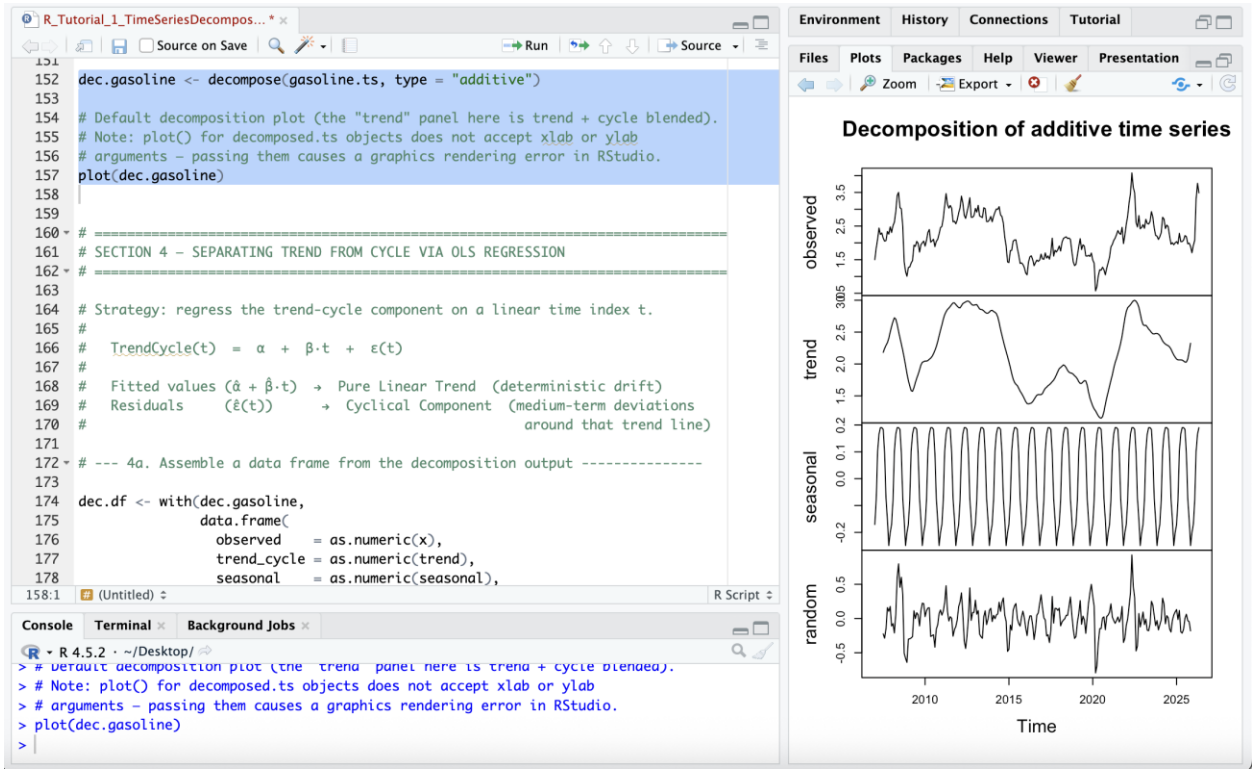
3. The data downloaded from Yahoo Finance is daily trading data; so we need to convert the daily closing prices into monthly data. The code uses the last available closing price of each month as the monthly value, so run the following code, and you will see:



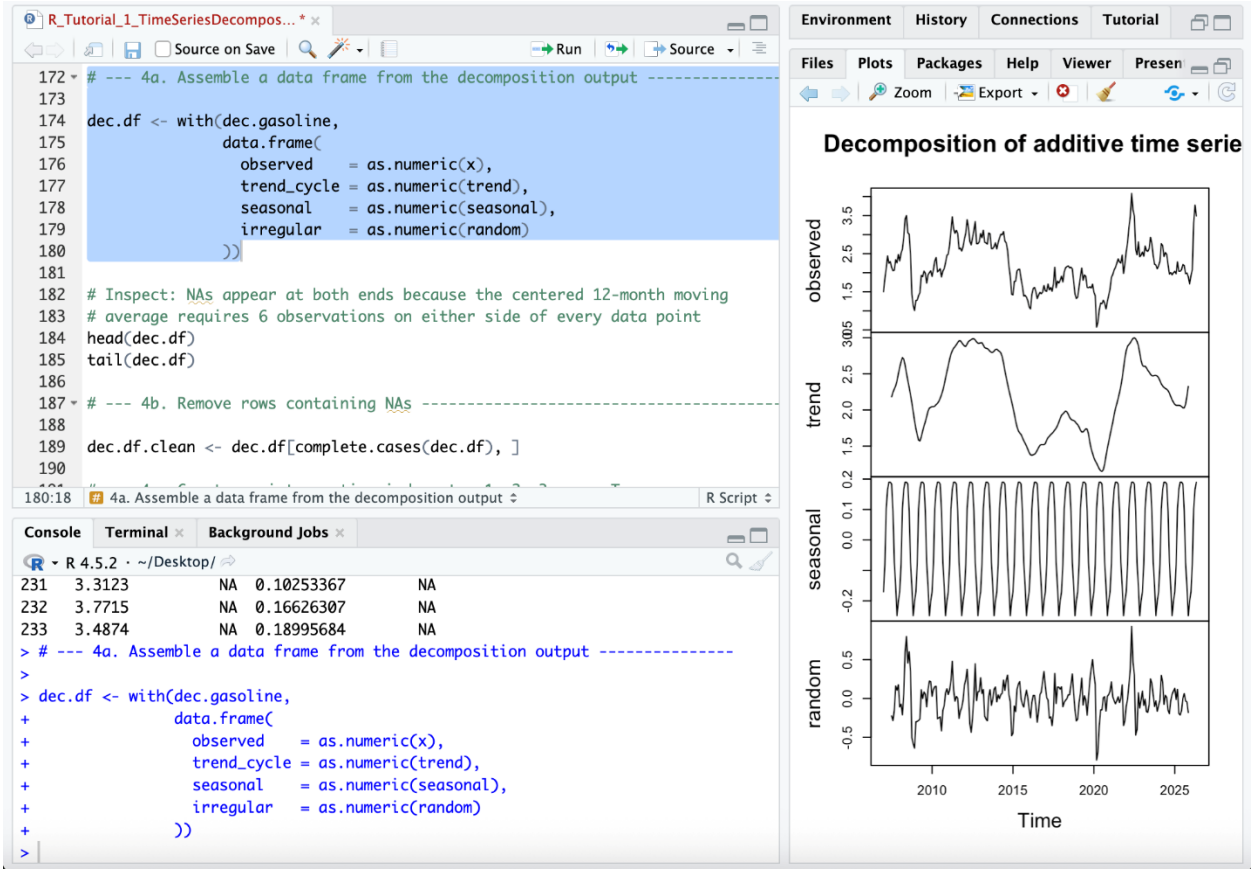
4. When we run the ‘plot’ function, we are able to see this line graph. However, based on our discussions in class, we cannot clearly separate the four components: trend, seasonal, cyclical, and random. Thus, we try to decompose the time series.



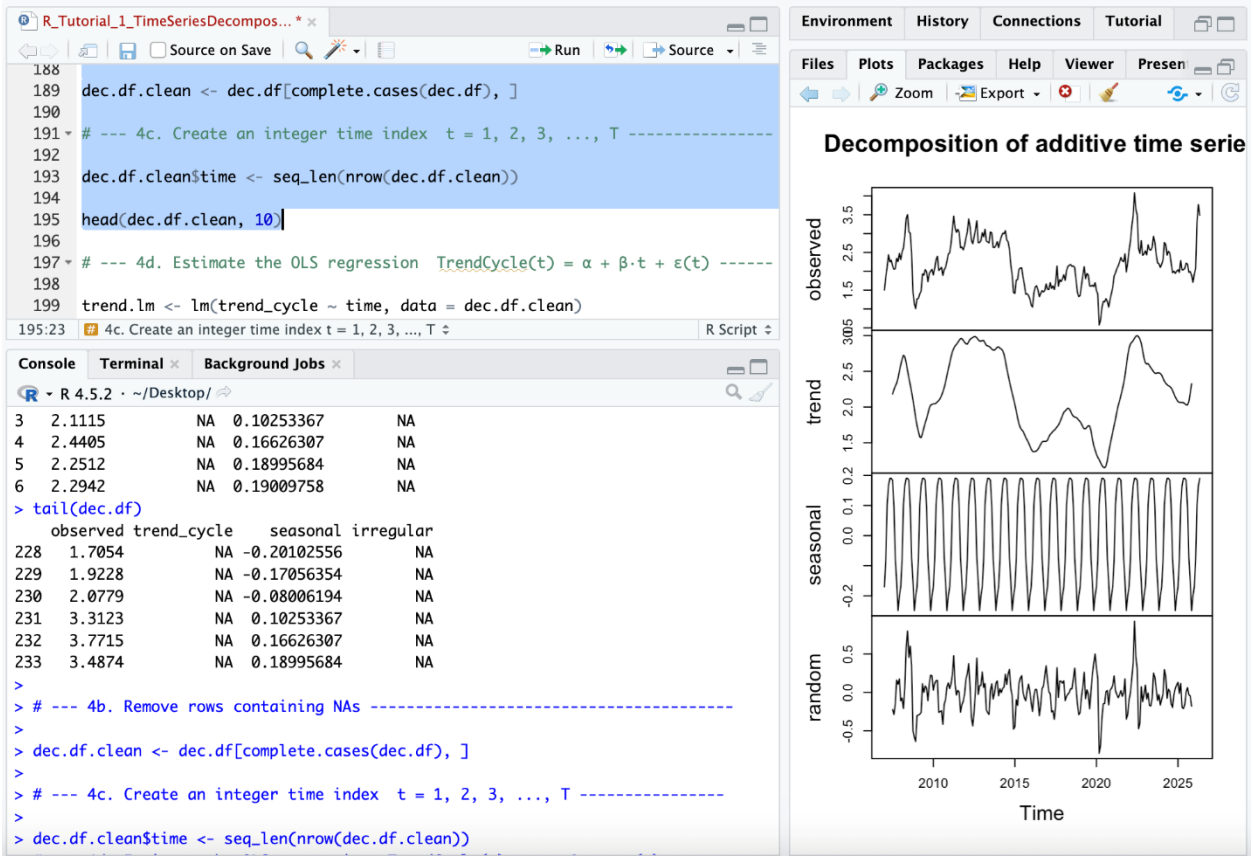
5. Then, let us try an additive decomposition. The R helps us decompose the time series into 3 components: trend, seasonal, and random. However, as discussed in class, we would like to have 4 components: linear trend, seasonal, cyclical, and random. We do the 4-component decomposition in later steps.



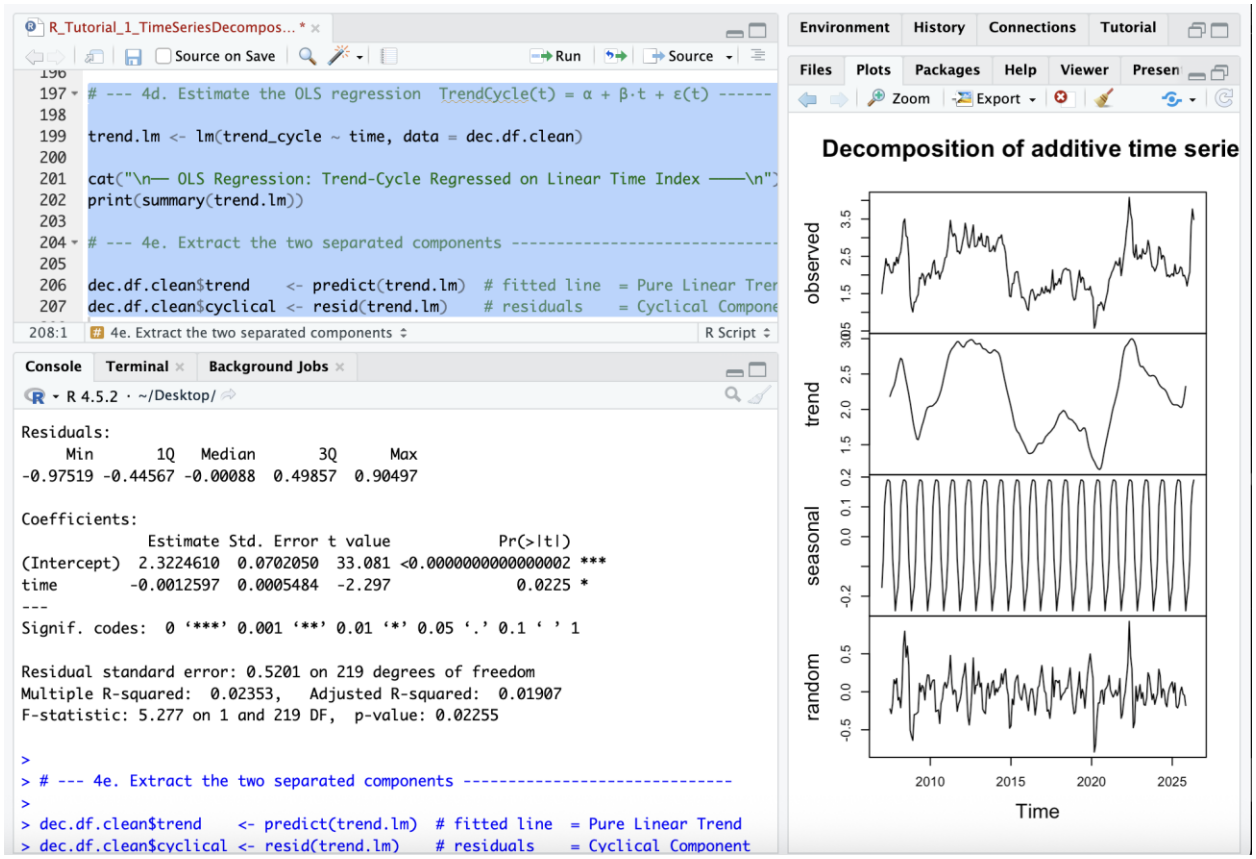
6. To decompose these 4 components without changing the original data, run the following code, and we create a new data frame called dec.df. In this data frame, the original trend component is stored as trend-cycle. In the later step, we will separate it into a pure linear trend and a cyclical component.



7. Before estimating the linear trend, we first need to inspect the decomposition data for missing values. We remove the missing values and create the time index, which will be used in the regression in the next step. Once you run the highlighted code, you will see the following:



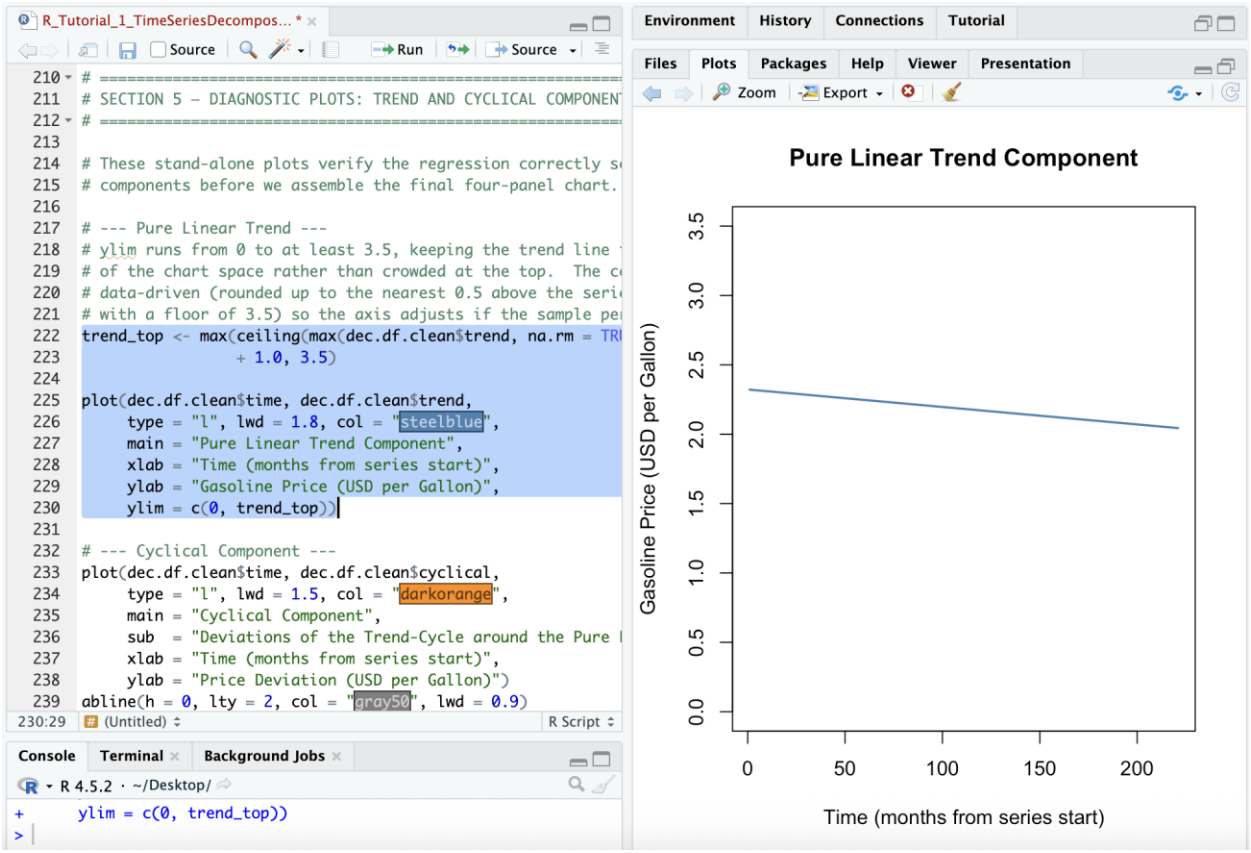
8. Now we can run the regression of the trend-cycle component on time to separate it into a pure linear trend and a cyclical component. After running the highlighted code, you should see the regression output below.



a. The fitted values represent the pure linear trend, and the residuals represent the cyclical component.

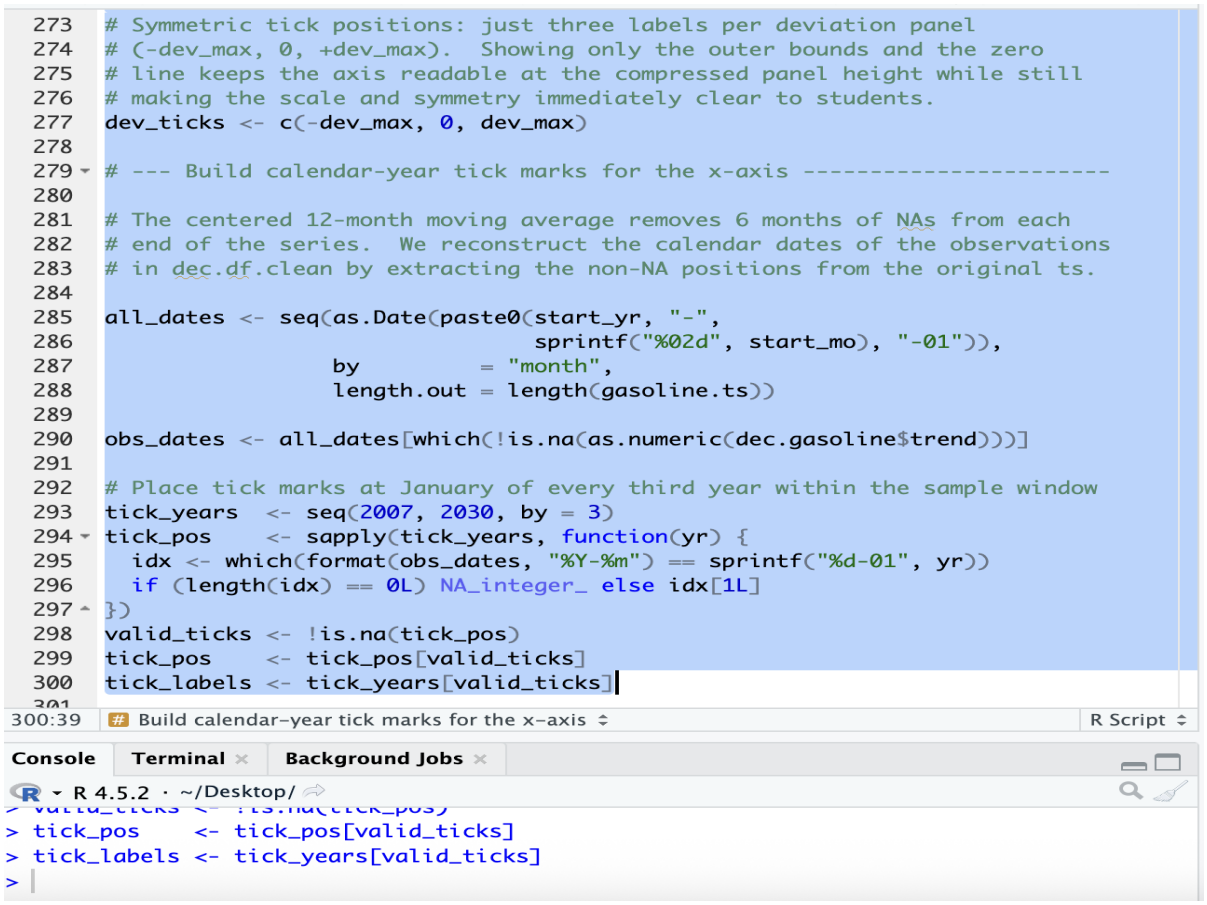
b. The regression model is stored as trend.lm. The fitted values are stored as dec.df.clean$trend and the residuals are stored as dec.df.clean$cyclical.

9. Then, if you run the plot function using the relevant data frame, you will see the following:



a. Note how we achieve a perfectly linear output in the graph, the estimated linear trend slopes slightly downward over time.

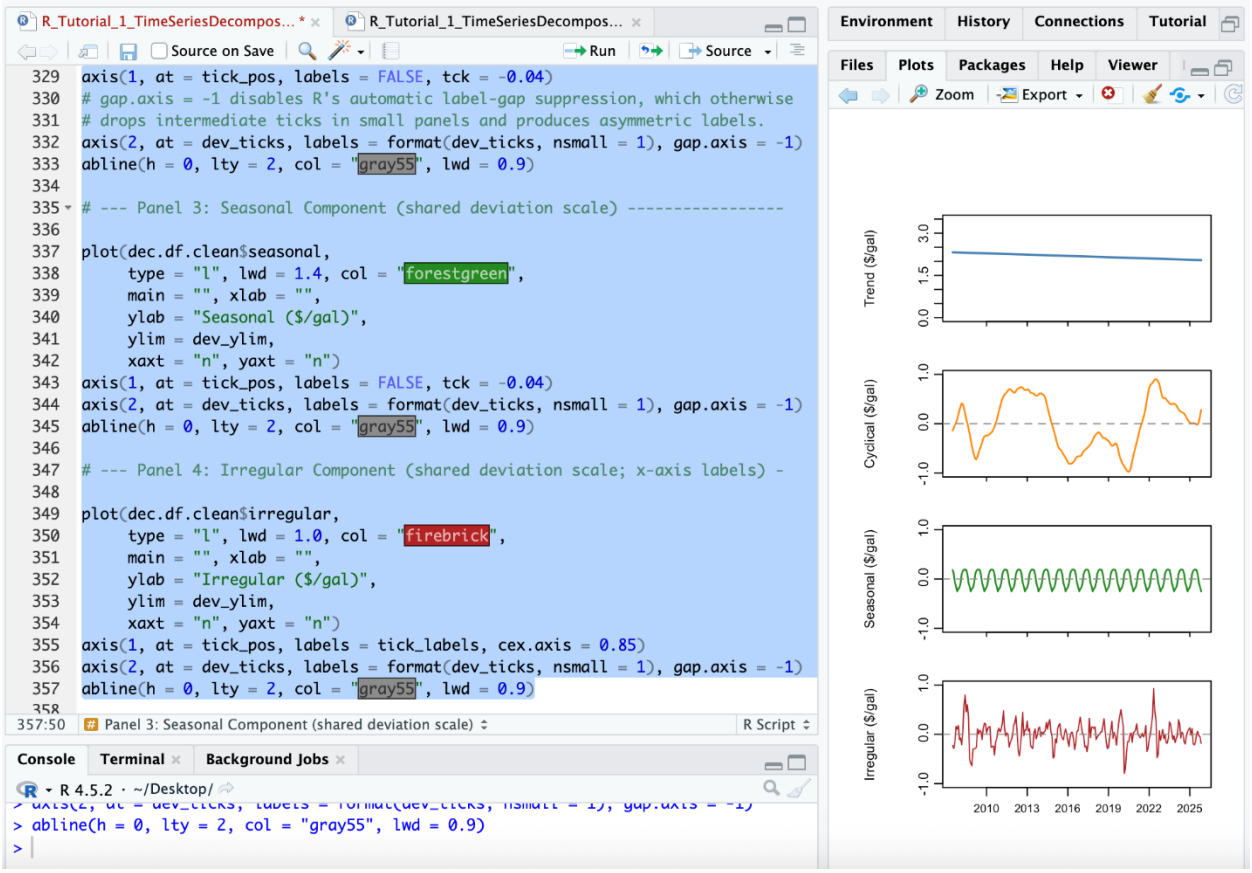
10. Before putting the four components into one graph, we need to first set up the axes. We use the same y-axis range for the cyclical, seasonal, and irregular components, so we can compare them easily. Run the following code:



11. Then, we combine all components into one graph panel. A few things to note:

    a. The pure linear trend slopes slightly downward over the sample period.

    b. We generate 4 components – linear trend, cyclical component, seasonal component, and irregular component. Based on the observations from the decomposition, you will note that our time series exhibits strong cyclicality. It is dominated by cyclicality.



The R code is provided along with the tutorial materials. Students should be able to run the code. The data is downloaded directly from Yahoo, so no data file is required. This R tutorial supplements class discussions on the decomposition topics and other decomposition examples discussed as part of the Module materials.